

Egypt Multi-Asset AI Portfolio Optimization System

Technical Architecture & Proposal

Prepared by: Mostafa Ibrahim

April 29, 2026

Abstract

This document outlines the architecture and methodology for an AI-driven, multi-asset portfolio optimization and prediction system focused on the Egyptian financial market. Integrating advanced machine learning pipelines and Modern Portfolio Theory (MPT), this project constructs a quantitative framework for autonomous asset allocation. The final algorithmic system will be deployed via a modern React.js user interface, providing a highly interactive and comprehensive quantitative asset management tool.

1 Introduction

In contemporary financial markets, optimal investment strategies necessitate empirical, data-driven methodologies. This project proposes a computational AI system to construct an optimal investment portfolio utilizing Egyptian financial assets.

The system synthesizes financial principles with advanced quantitative techniques, including machine learning classification, predictive regression, Monte Carlo simulations, and algorithmic optimization. The primary research question is: *How can capital be optimally allocated across diverse asset classes in Egypt to maximize returns while mitigating risk using Artificial Intelligence?*

2 Data Sourcing and Asset Allocation

A critical component of this quantitative system is the rigorous sourcing of high-fidelity financial data. The portfolio will be diversified across four primary asset classes:

Asset Class	Proxy / Instruments			Primary Data Source(s)
Equities	EGX30, EGX60	In-	Indices	Egyptian Exchange (EGX), Yahoo Finance API.
Fixed Income	Treasury Bills	(T-	Bills)	Central Bank of Egypt (CBE) statistical bulletins.
Real Estate	Property (Proxy)	Index		CAPMAS and Aqarmap Egypt Price Index.
Commodities	Gold (XAU/EGP)			World Gold Council (adjusted for EGP).

Table 1: Asset Classes and Explicit Data Sources

3 System Architecture and Visualization

The architecture follows a scalable data science pipeline, visualized below from data ingestion to the final React.js user interface.

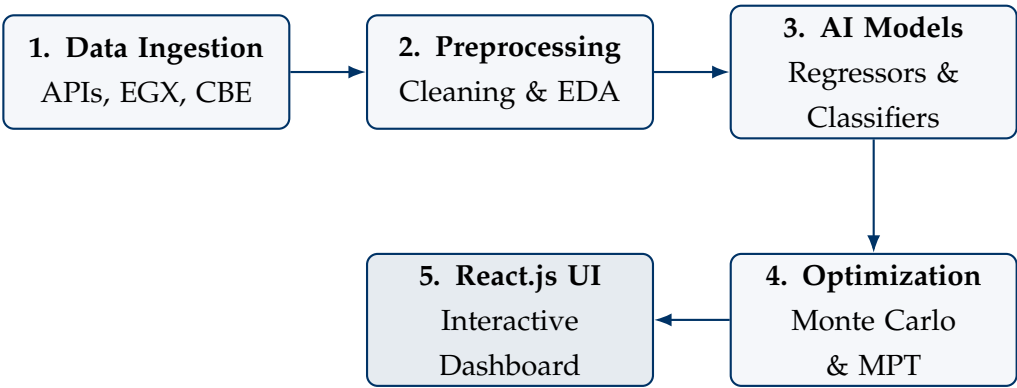


Figure 1: System Architecture and Data Pipeline

3.1 Exploratory Data Analysis (EDA)

EDA will uncover underlying market microstructures. The system will programmatically generate correlation matrices to ensure adequate mathematical diversification before optimization.

	Equities	T-Bills	Real Est.	Gold
Equities	1.00	-0.20	0.40	-0.45
T-Bills	-0.20	1.00	0.10	-0.15
Real Est.	0.40	0.10	1.00	0.25
Gold	-0.45	-0.15	0.25	1.00

Figure 2: Illustrative Asset Correlation Matrix (Red = Positive, Blue = Negative Correlation)

4 Predictive AI Modeling Pipeline

The system will deploy an ensemble of three specific regression models and three classification models.

4.1 Regression Models (Forecasting Continuous Returns)

1. **Multiple Linear Regression (MLR):** Statistical baseline for modeling historical lags against future returns.
2. **Random Forest Regressor:** Non-linear ensemble method to capture complex interactions and prevent overfitting.
3. **Support Vector Regressor (SVR):** Projects financial data into hyperplanes to predict price trajectories in high-dimensional spaces.

4.2 Classification Models (Predicting Market Regimes)

1. **Logistic Regression:** Probabilistic model estimating the likelihood of upward or downward market trends.

2. **Support Vector Machine (SVM):** Segregates market data into distinct regimes based on volatility and momentum.
3. **XGBoost Classifier:** A state-of-the-art boosting algorithm offering superior predictive accuracy for financial time-series.

5 Portfolio Optimization Theory

Rooted in Modern Portfolio Theory (MPT), the system will perform Monte Carlo simulations to mathematically plot the **Efficient Frontier**.

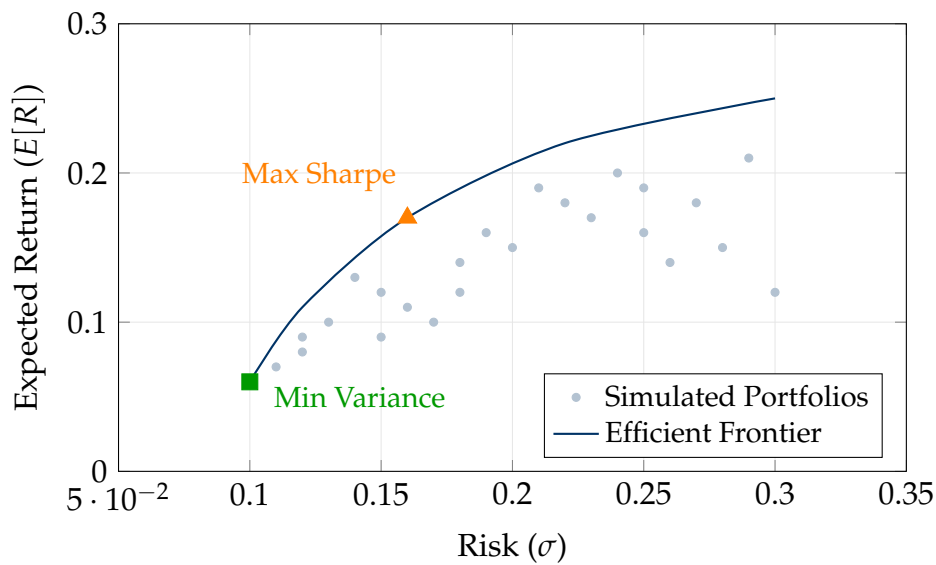


Figure 3: Simulated Efficient Frontier and Optimal AI Portfolios

To isolate optimal portfolios, the system calculates the Sharpe Ratio:

$$\text{Sharpe Ratio} = \frac{E[R_p] - R_f}{\sigma_p}$$

where $E[R_p]$ is portfolio return, R_f is the risk-free rate, and σ_p is portfolio standard deviation.

6 User Interface

Interactive React.js Application: A dynamic UI communicating via RESTful APIs with a Python backend, featuring interactive portfolio allocation and AI predictive model outputs.